# Lecture 37 — Why did the system behave that way?

**Fri Nov 20**

Last updated: October 8, 2026.

**Textbook:** Boyd & Vandenberghe, *VMLS* (PDF)

**Strang / MIT (optional):** Reading: Strang §6.1 (PDF) — eigenvalues
Video: 18.06 Lec 21 — eigenvalues and eigenvectors · MIT 18.06 playlist (Strang)

Reading: Strang §6.1 and Lachniet Ch. 5 (eigenvalues/eigenvectors). Related objectives: Lectures. Schedule.

Bring paper. Warm-up, fill-in notes, board examples, and **Your turn**.

## Learning objectives

By the end of class, you should be able to:

1. Compute eigenvalues and eigenvectors of a general matrix with software.
2. Verify $Av \approx \lambda v$ numerically.
3. Use dominant eigenbehavior to explain a simulated system.
4. Interpret complex eigenvalues as rotation/oscillation behavior.
5. Explore a symmetric-matrix or normal-mode example and observe orthogonal eigenvectors.

## Warm-up

Let

$$A = \begin{bmatrix} 1.1 & 0.2 \\ 0.1 & 0.8 \end{bmatrix}, \qquad v = \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

Hand-compute $Av$. Guess a plausible $\lambda$ so that $Av \approx \lambda v$. (Exact eigenpairs come from software in lab.)

## Fill in

**Software eigenpairs.** Octave `eig(A)` / Python `np.linalg.eig(A)` return eigenvalues and a matrix whose columns are eigenvectors.

**Verify.** For each reported pair $(\lambda, v)$, check

$$Av \approx _____$$

(componentwise; floating-point noise is expected). Never trust a bare printout without this check.

**Dominant behavior.** If $|\lambda_1| > |\lambda_2| \geq \cdots$ and $x_0$ has a component along the $\lambda_1$-eigenvector, then for large $t$

$$A^t x_0 \approx c\, \lambda_1^t\, v_1.$$

That explains Lab 4-style growth/decay/stabilization once you know the eigenpairs.

**Complex eigenvalues.** For a real matrix, nonreal eigenvalues come in conjugate pairs. In the real plane they produce _____ (rotation / oscillation), matching Monday's pure-rotation example.

**Normal modes (demo, not a DE course).** For a symmetric mass–spring style matrix, eigenvectors are often _____ (orthogonal). Software shows the modes; we do **not** derive differential equations today.

## Worked in class

**W1.** Software sketch. On the board / laptop, for $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$:

```
# Octave:  [V,D] = eig(A)     or    eig(A)
# Python:  vals, vecs = np.linalg.eig(A)
```

Read off $\lambda$'s and columns of $V$. Verify $Av_1 = \lambda_1 v_1$ for one column.

**W2.** Explain a simulation. Same $A$ as W1; $x_0 = (1, 0)$. Using eigenpairs, explain in one sentence why $x_t$ grows and which direction it approaches.

**W3.** Complex / oscillation. Software on $R = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ should report eigenvalues near $\pm i$. Connect that to the cycling you saw in Lab 4 stretch / Lecture 35.

**W4.** Symmetric demo. Let $S = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ (symmetric). Check that the two eigenvectors from W1 are orthogonal ($v_1^{\mathsf{T}} v_2 = 0$). Normal-mode pictures use the same idea in higher dimension.

## Your turn

**Y1.** Pseudocode: given $A$, obtain eigenpairs in software and write the one-line check that $Av \approx \lambda v$ for the first pair.

**Y2.** True or false: if $|\lambda_1| = 1.5$ and $|\lambda_2| = 0.5$, then for generic $x_0$ the long-run direction of $A^t x_0$ is the $\lambda_1$-eigendirection.

## Answers (Your turn)

Y1. Sample: `vals, V = np.linalg.eig(A)` then `A @ V[:,0] - vals[0]*V[:,0]` (should be near 0); Octave: `[V,D]=eig(A)` then `A*V(:,1) - D(1,1)*V(:,1)`.

Y2. **True.** The $\lambda_1$ term grows relative to $\lambda_2$, so the state aligns with $v_1$ (unless the $v_1$-component of $x_0$ is exactly zero).